LECTURE 27-28

29 November 2023 08:37

$$\iint_R x^2 y \, dy \, dx \qquad y = x^2 \quad y = \sqrt{x}$$

$$x = 0 \ , \ x = 1$$

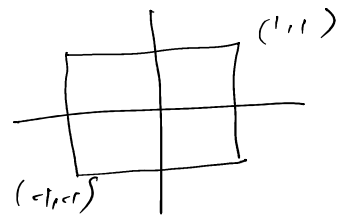
$$\int_0^1 \int_{\sqrt{x}}^{x^2} x^2 y \, dy \, dx = \int_0^1 x^2 \left(\frac{y^2}{2} \right)_{\sqrt{x}}^{x^2} dx$$

$$= \int_0^1 x^2 \cdot \left[\frac{x^4}{2} - \frac{x}{2} \right] dx$$

$$\iint_R (x^2 - y^2) \, dx \, dy$$

R : Rectangle $(-1,-1), (-1,1), (1,1) \ (1,-1)$

$$= \int_{-1}^{1} \left(\int_{-1}^{1} (x^2 - y^2) \, dx \right) dy$$



$$= \int_{-1}^{1} f(y) \, dy$$

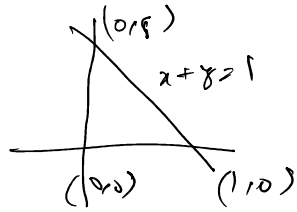
$$f(y) = \int_{-1}^{1} (x^2 - y^2) \, dx$$

$$= \left[\frac{x^3}{3} - y^2 x \right]_{-1}^{1}$$

Eg. $\int_0^1 x \sqrt{x^2+1} \, dx \qquad x^2 + 1 = t \qquad 2x \, dx = dt$

$$\int_1^2 \sqrt{t} \, \frac{dt}{2}$$

Eg $\iint_R e^{\frac{x-y}{x+y}} \, dx \, dy \qquad R : \{ x \geq 0, \ y \geq 0, \ x + y \leq 1 \}$



$x - y = u$

$x + y = v$

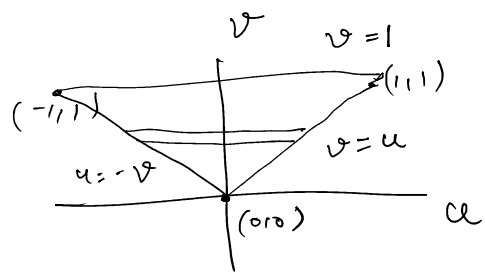
$$\iint_R f(x,y) \, dx \, dy = \iint_{R'} f(x(u,v), y(u,v)) \, |J(u,v)| \, du \, dv$$

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J(u,v,w) = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}_{3\times 3}$$

$$\left.\begin{aligned} x-y &= u \\ x+y &= v \end{aligned}\right\} \Rightarrow \begin{aligned} x &= \frac{v+u}{2} = \frac{v}{2}+\frac{u}{2} \\ y &= \frac{v-u}{2} = \frac{v}{2}-\frac{u}{2} \end{aligned} \qquad J(u,v) = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2}$$

$$\int_0^1 \int_{-v}^{v} e^{\frac{u}{v}} \cdot \frac{1}{2}\, du\, dv$$



$$\frac{1}{2}\int_0^1 \left(\int_{-v}^{v} e^{\frac{u}{v}}\, du\right) dv$$

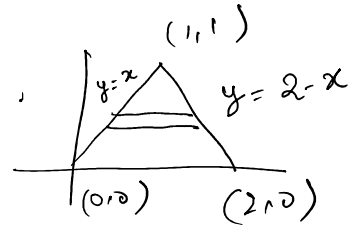
$$= \frac{1}{2}\int_0^1 \left[v e^{\frac{u}{v}} \right]_{-v}^{v} dv = \frac{1}{2}\int_0^1 v\left(e - e^{-1}\right) dv$$

$$= \frac{1}{2}\left(e - e^{-1}\right)\left[\frac{v^2}{2}\right]_0^1$$

$$= \frac{1}{4}\left(e - e^{-1}\right)$$

(2) $$\iint_R \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) dx\, dy$$

R:



$x+y$ $\quad x-y = u$

$(0,0)$ $(2,0)$

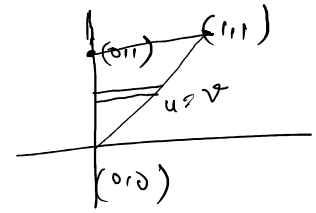
$x = u+v$ $\quad \frac{x-y}{2} = u$

$y = u-v$ $\quad \frac{x+y}{2} = v$

$$J(u,v) = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

$$\int_0^1 \int_0^v \sin v \cos u \; 2 \cdot du \, dv$$

$R' =$



$$\frac{1}{2} \iint_R 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) dx \, dy$$

$$\frac{1}{2} \int_0^1 \int_x^{2-x} \sin x + \sin y \quad dx \, dy$$

Polar co-ordinates : $x = r\cos\theta \quad y = r\sin\theta$

$$J(r,\theta) = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

$$\iint_R f(x,y) \, dx \, dy = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r,\theta) \, r \, dr \, d\theta$$

① Find volume of sphere with radius 'a'.

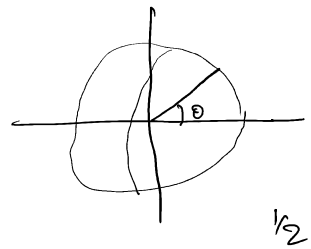
$$x^2 + y^2 + z^2 = a^2$$

$$f(x,y) = \sqrt{a^2 - x^2 - y^2}$$

$$\int_0^{2\pi} \int_0^a \sqrt{a^2 - r^2} \cdot r \, dr \, d\theta \qquad a^2 - r^2 = t$$

$$\int_0^{2\pi}\int_0^{a} \sqrt{a^2-r^2}\cdot r\,dr\,d\theta \qquad a^2-r^2=t$$

$$-2r\,dr = dt$$



$$\int_0^{2\pi}\int_{a^2}^{0} \sqrt{t}\,\frac{dt}{-2}\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{a^2} t^{1/2}\,dt\,d\theta$$

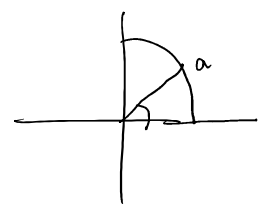
$$= \frac{1}{2}\int_0^{2\pi}\left(\frac{2}{3}t^{3/2}\right)_0^{a^2} d\theta$$

$$= \frac{1}{2}\cdot\frac{2}{3}\int_0^{2\pi} a^3\,d\theta = \frac{a^3}{3}\int_0^{2\pi} d\theta$$

$$= \frac{2\pi a^3}{3}$$

2) $f(x,y) = \dfrac{1}{x^2+y^2+1}$ R: First quadrant circle with radius a

$$\iint_R f(x,y)\,dx\,dy = \int_0^{\pi/2}\int_0^{a} \frac{1}{r^2+1}\,r\,dr\,d\theta$$



$$= \frac{1}{2}\int_0^{\pi/2}\int_1^{a^2+1} \frac{1}{t}\,dt\,d\theta \qquad r^2+1=t$$

$$2r\,dr = dt$$

$$= \frac{1}{2}\int_0^{\pi/2} (\log t)_1^{a^2+1}\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} \log(a^2+1)\,d\theta$$

$$= \frac{1}{2}\log(a^2+1)\int_0^{\pi/2} d\theta$$

$$= \frac{\pi}{4} \log (a^2+1)$$

Volume between two surfaces $f(x,y)$ & $g(x,y)$

$$\iint_R f(x,y) - g(x,y) \, dx \, dy \qquad f(x,y) \geq g(x,y)$$

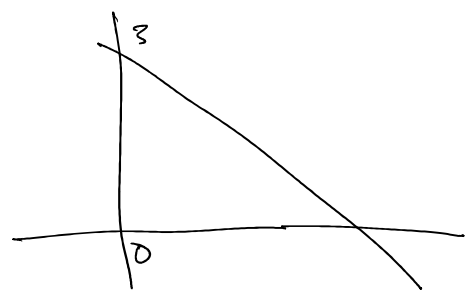
Eg: $2x + 3y - z = 8$ & $x + 3y + z = 10$

$$\int_{y=0}^{y=3} \int_{x=0}^{x=6-2y} (2x+3y-8) - (10-x-3y) \, dx \, dy$$

$$2x+3y-8 = 10-x-3y$$

$$3x+6y = 18 \Rightarrow x+2y = 6$$

$$\int_0^3 \int_0^{6-2y} 3x+6y-18 \, dx \, dy = 54$$



$$\int_0^3 \int_0^{6-2y} \left(\int_{10-x-2y}^{2x+3y-8} 1 \right) dz \, dx \, dy$$

Triple integration